

Strain Based Analysis of Multi Hole Interaction in Acrylic Specimens using Digital Image Correlation

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Abstract

This project investigates how multiple circular holes interact to alter the local strain field in acrylic specimens loaded in tension, using Digital Image Correlation (DIC) as the primary measurement tool. Eight specimen configurations were tested: a no hole baseline (NH), a single hole (1H), two- and three hole collinear arrangements at normalised spacings $s/d = 2$ and $s/d = 4$ (2H2, 2H4, 3H2, 3H4), and three hole staggered arrangements at the same two spacings (3H2S, 3H4S). Average strains were recorded using MatchID over ROIs placed on annular regions around each hole, ligaments between holes, and a far field region. Two normalised metrics were used: the strain concentration factor K_ε and the interaction index I , both referenced to each specimen's own far field strain. The data show that ligament interaction strengthens significantly with closer hole spacing, that the middle hole in a collinear three hole array at $s/d = 2$ is shielded by its outer neighbours, and that staggering the centre hole laterally reduces vertical ligament loading by about 24%. The results align with classical hole interaction theory and demonstrate the value of DIC for resolving local strain redistribution in perforated structures.

1 Overview

(a) Problem Statement. Holes in a loaded structural component create localised strain amplification that can drive crack initiation well before the nominal stress reaches the material limit. When multiple holes are present, they interact, altering the strain in the ligament between them and modifying the concentration around each hole individually. The primary goal of this project is to experimentally quantify, using DIC, how hole count, spacing, and arrangement (collinear vs. staggered) affect local strain distributions in acrylic specimens.

(b) Steps Executed. (i) Acrylic specimens were prepared in eight configurations (hole count: 0, 1, 2, 3; spacing $s/d \in \{2, 4\}$; arrangement: collinear or staggered). (ii) A random speckle pattern was applied to each specimen for DIC tracking. (iii) Each specimen was loaded in tension on a universal testing machine while images were captured continuously. (iv) MatchID was used to process the image sequences; ROIs were defined manually and average strains were recorded for each zone. (v) ROIs were placed over annular zones around holes, ligaments between holes, and far field zones. (vi) Normalised metrics K_ε and I were computed and compared across configurations.

(c) Expected Outcome. A clear demonstration that ligament strain interaction increases as holes move closer together, that annular concentration varies with hole count and arrangement, and that staggering the centre hole reduces the load carried by the primary vertical ligament relative to a fully collinear array.

(d) Actual Outcome and Key Findings. All three expected trends were observed. The three hole collinear case showed a particularly pronounced interaction effect: normalised ligament strain nearly doubled when spacing decreased from $s/d = 4$ to $s/d = 2$ ($I : 0.395 \rightarrow 0.757$). A notable additional finding was middle hole shielding in the collinear array at $s/d = 2$: the middle hole showed lower annular concentration ($K_\varepsilon = 0.711$) than the outer holes ($K_\varepsilon = 1.238$), consistent with classical analytical predictions. The staggered configuration reduced vertical ligament interaction by about 24%

compared with the collinear case at the same spacing.

(e) Most Innovative Aspect. The study demonstrates how simple geometric changes, such as lateral offsetting of a hole, can redistribute load paths and significantly reduce ligament strain, offering practical insight for the design of perforated components.

(f) Future Work. (i) Four hole specimens were fabricated and tested in this project, but DIC analysis could not be performed due to unavailability of the analysis setup at the time of submission. Completing this would extend the study to a fourth hole count and reveal whether shielding and ligament loading continue to scale with hole number. (ii) Testing additional configurations such as varying offset, nonsymmetric arrangements, or intermediate s/d values would map the interaction landscape more completely. (iii) Multiple specimens per configuration are needed to separate genuine geometric effects from experimental error. (iv) Finite element simulations could provide a quantitative check on the experimental trends.

(g) Who Did What.

- Nikita Rewad: specimen preparation, speckle patterning, and mechanical testing
- Harshita Jaronde: mechanical testing, image acquisition, and literature review
- Aayush Shah: DIC post processing, ROI extraction, and report writing
- Shrenik Jain: specimen preparation, speckle patterning, and report writing
- All members contributed to discussion and interpretation of results

2 Materials and Methods

(a) Materials. The following materials were used:

- Cast acrylic (PMMA) sheet, 4 mm thickness. Specimens were laser cut at the Microfactory Facility in IIT Bombay, to the ASTM D638 Type I dog bone geometry (overall length 165 mm, gauge length 50 mm, gauge width 13 mm, grip section width 19 mm, fillet radius 76 mm). Holes of the required diameter were incorporated directly into the laser cut geometry at the prescribed centre to centre spacings to achieve the target s/d ratios.
- White spray paint (speckle base coat)
- Black marker pen (speckle pattern)

(b) Methods. Each specimen was loaded uniaxially in tension under displacement control. Images were captured at a fixed frame rate throughout the test. The image sequence was processed using MatchID, the DIC software used in this study. Full field strain maps were not available in MatchID at the subset level for these specimens, so instead, regions of interest (ROIs) were defined manually over the relevant zones and the average strain within each ROI was recorded directly. ROIs were placed over annular zones around each hole (A1, A2), ligaments between adjacent holes (vertical L1 and diagonal L2 where applicable), and far field zones (FF). Two normalised quantities were computed: $K_\epsilon = \epsilon_{\text{ann}}/\epsilon_{\text{FF}}$ and $I = \epsilon_{\text{lig}}/\epsilon_{\text{FF}}$, both referenced to the same specimen’s far field strain to eliminate load level differences across specimens.

(c) Equipment and Location. Tensile testing was performed on a universal testing machine (UTM). Imaging used a camera with two continuous light sources illuminating the specimen surface. All testing was carried out in the Solid Mechanics Lab, Department of Mechanical Engineering.

3 Results and Discussion

Figure 1 shows the specimen layout used in the DIC setup alongside a representative strain map from MatchID. DIC measured strain values and derived normalised indices for all eight specimens are summarised in Table 1. The no hole specimen (NH) gives $\epsilon_{\text{FF}} = 0.012022$ as a global reference, though all K_ϵ and I calculations use each specimen’s own FF strain. For the single hole specimen, $K_\epsilon = 1.48$,

which is below the classical theoretical value of 3. This is expected: the annular ROI spans a wide band around the hole and the DIC averaged strain is pulled down by the lower strain material away from the hole edge. The classical peak of 3 would only be recovered by sampling right at the hole boundary with a very narrow ROI.

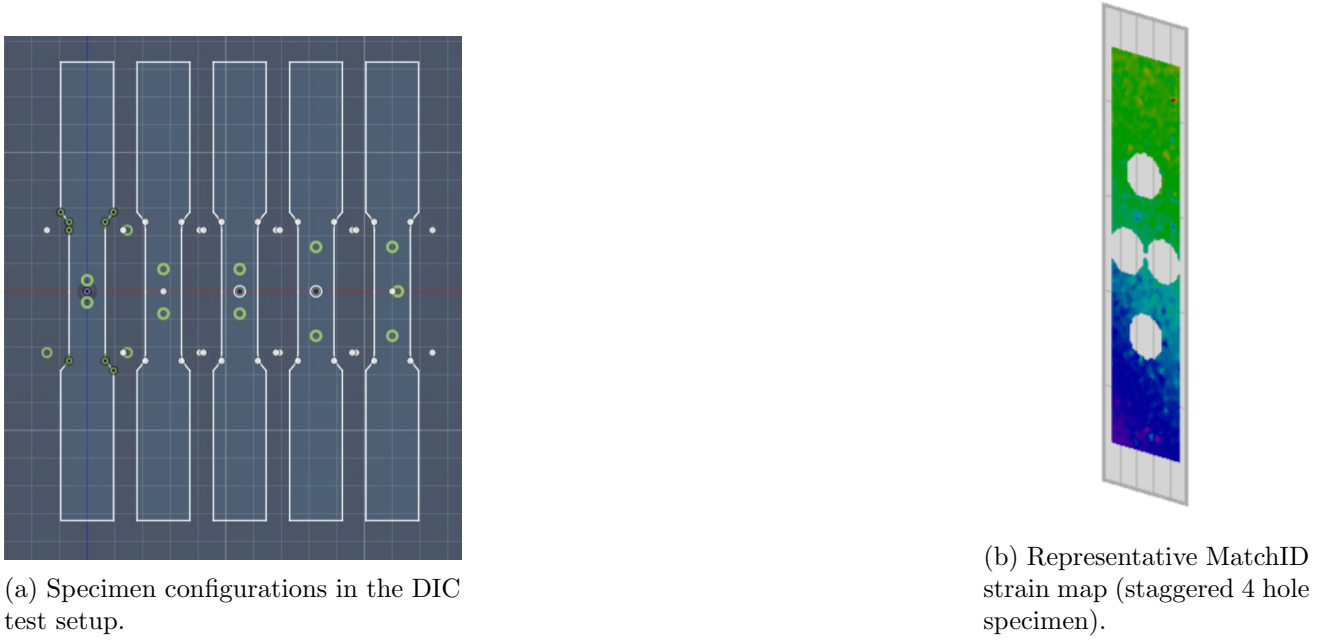


Figure 1: Experimental setup and DIC output.

Table 1: DIC measured strain values and derived indices. All K_ϵ and I values are normalised by each specimen’s own ϵ_{FF} . The A2 value for 3H2S is excluded.

Specimen	s/d	ϵ_{FF}	$K_{\epsilon,A1}$	$K_{\epsilon,A2}$	I_{vert}	I_{diag}
NH	–	0.012022	–	–	–	–
1H	–	0.006701	1.480	–	–	–
2H2	2	0.004099	1.166	–	0.862	–
2H4	4	0.005808	1.428	–	0.663	–
3H2	2	0.002639	1.238	0.711	0.757	–
3H4	4	0.003550	1.576	2.024	0.395	–
3H2S	2	0.003348	0.918	–	0.578	0.178
3H4S	4	0.000661	1.269	1.982	–	1.401

The A2 reading for 3H2S was 0.123, which is non physical for an elastic tensile test and is excluded. It most likely reflects DIC tracking failure or crack initiation at the offset hole at that frame.

Effect of spacing on ligament interaction. For both the two hole and three hole collinear configurations, I decreases as spacing increases, confirming that hole to hole interaction weakens at larger separations (Table 2). The two hole case shows a moderate reduction of about 23% going from $s/d = 2$ to $s/d = 4$ ($I : 0.862 \rightarrow 0.663$). The three hole collinear case shows a much sharper drop of roughly 48% ($I : 0.757 \rightarrow 0.395$). This suggests that interaction between holes scales nonlinearly with hole count: adding a third collinear hole amplifies the sensitivity of the ligament to spacing, likely because the middle hole is simultaneously interacting with two neighbours rather than one. This is the clearest

and most consistent trend in the dataset.

Table 2: Effect of spacing on the normalised ligament interaction index I .

Configuration	I at $s/d = 2$	I at $s/d = 4$	Reduction
Two hole collinear (vert. lig.)	0.862	0.663	$\sim 23\%$
Three hole collinear (vert. lig.)	0.757	0.395	$\sim 48\%$

Annular concentration and middle hole shielding. Table 3 summarises the annular concentration factors for the three hole collinear array. At $s/d = 2$, the middle hole shows $K_{\varepsilon, \text{mid}} = 0.711$, which is noticeably lower than the outer hole value of $K_{\varepsilon, \text{top}} = 1.238$. This is the opposite of what one might naively expect, since the middle hole has two neighbours interacting with it. The explanation lies in load path shielding: in a collinear array loaded along the hole axis, the outer holes intercept the incoming load first and partially relieve the stress around the middle hole’s annular zone. The outer holes, sitting at the ends of the chain with undisturbed far field material on one side, concentrate more strain in their annular regions. This behaviour has been predicted analytically for axially aligned hole arrays and is consistent with what the data shows here.

At $s/d = 4$, the picture reverses. The middle hole now shows a higher concentration ($K_{\varepsilon, \text{mid}} = 2.024$) than the outer hole ($K_{\varepsilon, \text{top}} = 1.576$). One interpretation is that at larger spacing, the holes begin to behave more independently and the shielding interaction from the outer holes is too weak to suppress the bilateral concentration at the middle hole. However, since only one specimen was tested per configuration, it is also possible that this reversal reflects experimental scatter rather than a genuine physical phenomena, and it requires confirmation with repeat tests.

Table 3: Annular concentration factors for the three hole collinear array.

Spacing	$K_{\varepsilon, \text{top}}$	$K_{\varepsilon, \text{mid}}$	Behaviour
$s/d = 2$	1.238	0.711	Middle hole shielded
$s/d = 4$	1.576	2.024	Shielding lost

Staggered configuration. Offsetting the centre hole laterally creates two distinct ligaments: a vertical one (L1) between the top and bottom holes, and a diagonal one (L2) between the top hole and the offset centre hole. At $s/d = 2$, the vertical ligament carries $I_{\text{vert}} = 0.578$ while the diagonal carries only $I_{\text{diag}} = 0.178$ (Table 4). This is physically expected since the primary load path runs along the loading axis and the vertical bridge between the two axially aligned holes bears most of the load, while the inclined bridge to the offset hole is comparatively lightly loaded.

Comparing the staggered and collinear configurations at $s/d = 2$, the vertical ligament interaction drops from $I = 0.757$ (3H2) to $I = 0.578$ (3H2S), a reduction of about 24%. This shows that simply offsetting the centre hole laterally partially redirects the load away from the primary vertical path, which could be useful for design applications where reducing peak ligament strain is desirable without removing a hole entirely.

Sources of Error.

DIC subset size and speckle quality. The speckle pattern was applied by hand using a black marker over white spray paint, which produced non uniform speckle sizes and density. We attempted to reduce the subset size and use smaller ROIs to better capture the steep strain gradients near hole edges. However, below a certain threshold the correlation became highly unstable because individual subsets no longer contained sufficient unique speckle information. This imposed a hard lower limit on

Table 4: Staggered configuration: ligament comparisons at $s/d = 2$.

Comparison ($s/d = 2$)	Value 1	Value 2
Vertical lig. vs. diagonal lig. (3H2S)	$I_{\text{vert}} = 0.578$	$I_{\text{diag}} = 0.178$
Staggered vs. collinear (vert. lig.)	$I^{(3H2S)} = 0.578$	$I^{(3H2)} = 0.757$

how close to the hole boundary the annular ROIs could be placed, and is the primary reason K_ϵ values are systematically well below the theoretical peak of 3. The non physical A2 reading for 3H2S is also likely a consequence of this instability at a frame near crack initiation where the correlation degrades rapidly.

Clamping and alignment. Manual clamping means there is always some possibility of slight misalignment between the specimen axis and the loading axis. This introduces bending and produces non uniform far field strains across the width of the specimen. It cannot be fully ruled out and may contribute to the asymmetry seen between A1 and A2 readings in configurations that are nominally symmetric about the midplane.

Lighting variations. Digital Image Correlation assumes consistent illumination between the reference and deformed images. In practice, small variations in lighting intensity or direction can occur due to ambient conditions or slight changes in setup during testing. These variations alter the grayscale intensity patterns used for correlation, reducing tracking accuracy and introducing noise into the displacement and strain fields. The effect is most pronounced in regions with high strain gradients, where accurate subset matching is already challenging.

Frame selection. Each specimen was analysed at a different frame, meaning the DIC strain fields correspond to different applied loads. Per specimen FF normalisation removes most of this effect but cannot compensate fully for cases where the selected frame is close to or past the onset of localised damage, at which point the DIC strains near the hole boundary are no longer representative of an elastic response.

4 Conclusions

- Ligament interaction strengthens significantly with closer hole spacing: in the three hole collinear array, normalised ligament strain nearly doubles from $I = 0.395$ at $s/d = 4$ to $I = 0.757$ at $s/d = 2$.
- The middle hole in a collinear three hole array at $s/d = 2$ is shielded by its outer neighbours ($K_\epsilon = 0.711$), exhibiting lower annular concentration than the outer holes ($K_\epsilon = 1.238$), consistent with classical analytical predictions for axially loaded collinear hole arrays.
- Staggering the centre hole laterally reduces vertical ligament interaction by $\sim 24\%$ relative to the collinear case at the same spacing; the diagonal ligament carries only about one third of the vertical ligament's normalised strain.
- ROI averaged DIC underestimates classical peak concentrations. Attempts to reduce subset size and ROI width were limited by speckle quality, and finer, more uniform speckle patterns are needed in future tests to approach the theoretical $K = 3$.
- Four hole specimens were fabricated and mechanically tested but could not be analysed by DIC due to unavailability of the setup at the time of submission; completing this analysis is the most immediate next step and would help us verify our trends.